**NOTE TO APPLICANTS: IT IS IMPORTANT FOR YOU TO INCLUDE ALL RELEVANT INFORMATION ABOUT YOUR RESEARCH IN THIS APPLICATION FORM AS YOUR ETHICAL APPROVAL WILL BE BASED ON THIS FORM. THEREFORE ANYTHING NOT INCLUDED WILL NOT BE PART OF ANY ETHICAL APPROVAL.**

**YOU SHOULD READ THE ETHICS APPLICATION GUIDELINES AND HAVE THEM AVAILABLE AS YOU COMPLETE THIS FORM.**

APPLICATION FORM

## SECTION A APPLICATION FOR ETHICAL REVIEW: HIGH RISK

**A1**

Isolation of peripheral blood cells from healthy donors to investigate immune responses to infectious diseases

| | |
|---|---|
| Date of Submission: 28/4/22 | Proposed Data Collection Start Date: 1/6/2022 |
| UCL Ethics Project ID Number: 22589/001 | Proposed Data Collection End Date: 31/5/2026 |

**Is this application for continuation of a research project that already has ethical approval?** ***For example, a preliminary/pilot study has been completed and this is an application for a follow-up project? If yes, please provide the information requested below.***

Project ID for the previous study: N/A

**A2**

### Principal Researcher

*Please note that a student – undergraduate, postgraduate or research postgraduate cannot be the Principal Researcher for Ethics purposes.*

| | |
|---|---|
| Full Name: Professor Mahdad Noursadeghi | Position Held: Professor of Infectious Diseases |
| Name and Address of Department: Division of Infection and Immunity, UCL Cruciform Building Gower Street, London, WC1E 6BT | Email: m.noursadeghi@ucl.ac.uk |
| | Telephone: 02031082128 |
| | Fax: Not applicable |

### Declaration To be Signed by the Principal Researcher

- I have met with and advised the student on the ethical aspects of this project design *(applicable only if the Principal Researcher is not also the Applicant).*
- I understand that it is a UCL requirement for both students & staff researchers to undergo Disclosure and Barring Service (DBS) Checks when working in controlled or regulated activity with children, young people or vulnerable adults. The required DBS Check Disclosure Number(s) is: not applicable
- I have obtained approval from the UCL Data Protection Officer stating that the research project is compliant with the General Data Protection Regulation 2018. My Data Protection Registration Number is: **Z6364106/2022/04/105 medical research**
- I am satisfied that the research complies with current professional, departmental and university guidelines including UCL's Risk Assessment Procedures and insurance arrangements.
- I undertake to complete and submit the 'Continuing Review Approval Form' on an annual basis to the UCL Research Ethics Committee.
- I will ensure that changes in approved research protocols are reported promptly and are not initiated without approval by the UCL Research Ethics Committee, except when necessary to eliminate apparent immediate hazards to the participant.
- I will ensure that all adverse or unforeseen problems arising from the research project are reported in a timely fashion to the UCL Research Ethics Committee.
- I will undertake to provide notification when the study is complete and if it fails to start or is abandoned.

**SIGNATURE:** **DATE:** 25/4/22

| A3 | **Applicant(s) Details** *(if Applicant is not the Principal Researcher e.g. student details):* add Jacky & Bahel | |
|---|---|---|
| | Full Name: Clare Thakker | |
| | Position Held *i.e.undergraduate/bachelor or masters project (if so, provide course title/number, PhD, staff led research project which may involve one or more students*: Wellcome Trust Clinical Training Fellow | |
| | Name and Address of Department: Division of Infection and Immunity, UCL Cruciform Building Gower Street, London, WC1E 6BT | Email: c.thakker@ucl.ac.uk<br>Telephone: NA<br>Fax: NA |
| | Full Name: Adlina Chan | |
| | Position Held: Research assistant | |
| | Name and Address of Department: Division of Infection and Immunity, UCL Cruciform Building Gower Street, London, WC1E 6BT | Email: adlina.chan.17@ucl.ac.uk<br>Telephone: NA<br>Fax: NA |
| | Full Name: Robert Gray | |
| | Position Held: NIHR Academic Clinical Lecturer | |
| | Name and Address of Department: Division of Infection and Immunity, UCL Cruciform Building Gower Street, London, WC1E 6BT | Email: rekgrgr@ucl.ac.uk<br>Telephone: NA<br>Fax: NA |
| | Full Name: Gabriele Pollara | |
| | Position Held: Postdoctoral researcher | |
| | Name and Address of Department: Division of Infection and Immunity, UCL Cruciform Building Gower Street, London, WC1E 6BT | Email: g.pollara@ucl.ac.uk<br>Telephone: NA<br>Fax: NA |
| | Full Name: Lucy Bell | |
| | Position Held: NIHR Academic Clinical Lecturer | |
| | Name and Address of Department: Division of Infection and Immunity, UCL Cruciform Building Gower Street, London, WC1E 6BT | Email: lucy.bell.09@ucl.ac.uk<br>Telephone: NA<br>Fax: NA |

## A4 Sponsor/ Other Organisations Involved and Funding

a) **Sponsor:** ☒ **UCL** ☐ **Other institution**

If your project is sponsored by an institution other than UCL please provide details: N/A

b) **Other Organisations**: If your study involves another organisation, please provide details. *Evidence that the relevant authority has given permission should be attached or confirmation provided that this will be available upon request.* N/A

c) **Funding:** What are the sources of funding for this study and will the study result in financial payment or payment in kind to the department or College? *If study is funded solely by UCL this should be stated, the section should not be left blank.*

Funding source: Wellcome Trust

- Wellcome Trust Investigator Award to Professor Mahdad Noursadeghi (Wellcome Ref: 207511/Z/17/Z, UCL Project No: 542477)
- Wellcome Trust Clinical Fellowship Award to Dr Clare Thakker (Wellcome Ref: 225477/Z/22/Z, UCL Project No: 566704)

| A5 | **Signature of Head of Department [or Chair of your Departmental Research Ethics Committee/Departmental Ethics Lead]**<br>*(This must not be the same signature as the Principal Researcher)*<br><br>**A. I have discussed this project with the principal researcher who is suitably qualified to carry out this research and I approve it.**<br><br>**I am satisfied that *[please highlight as appropriate]*:**<br><br>**(1) Data Protection registration:**<br>• **has been satisfactorily completed**<br>• **has been initiated**<br>• **is not required**<br><br>**(2) a risk assessment:**<br>• **has been satisfactorily completed**<br>• **has been initiated**<br><br>**(3) appropriate insurance arrangements are in place and appropriate sponsorship [funding] has been approved and is in place to complete the study.** ☒ **Yes** ☐ **No**<br><br>**(4) a Disclosure and Barring Service check(s):**<br>• **has been satisfactorily completed**<br>• **has been initiated**<br>• **is not required**<br><br>*Links to details of UCL's policies on the above can be found at: http://ethics.grad.ucl.ac.uk/procedures.php*<br><br>****If any of the above checks are not required please clarify why below.**<br>**Not applicable** |
|---|---|

**PRINT NAME:** Professor Emma Morris

**SIGNATURE:** **DATE:** 27.04.2022

## SECTION B DETAILS OF THE PROJECT

****It is essential that Sections B1 and B2 are completed in simple understandable lay language that a non-expert could understand or you risk your project being rejected**

| B1 | **Please provide a brief summary of the project in simple lay person's prose outlining the intended value of the project, giving necessary scientific background.** *(max 500 words)*.<br><br>Infectious diseases remain a major cause of death and disability worldwide. We study how the human immune system interacts with the microbial organisms that cause a number of infectious diseases including tuberculosis and Covid-19. Through our broad programme of work we aim to enhance our mechanistic understanding of normal immune cell function, how this is affected in the context of infectious diseases and how this can contribute to immune protection and pathogenesis in infectious diseases. Our work has the potential to identify novel targets for treatment, diagnosis and prognostication in infectious diseases. |
|---|---|

To address these aims, we undertake laboratory research using *in vitro* experimental models. These *in vitro* models use human immune cells obtained from peripheral blood samples from healthy volunteers. These cells are used in laboratory experiments to investigate the factors which make them more or less susceptible to infection, their ability to restrict pathogen growth/replication, and how they react to or are otherwise affected by infection.

**B2**

**Briefly characterise in simple lay person's prose the research protocol, type of procedure and/or research methodology (e.g. observational, survey research, experimental). Give details of any samples or measurements to be taken** *(max 500 words).*

We require healthy volunteers to donate blood, from which immune cells can be isolated for use in our laboratory research.

Healthy adults, aged 18-65, will be recruited via advertisements e.g. posters, advertisements in UCL email newsletters, flyers at UCL occupational health screening and advertisement on the research team's website (**www.innate2adaptive.uk/volunteer**). Participants who self-identify and contact the study team (via details provided in advertisements or the form on the group website) will be asked to give written informed consent to donate a maximum volume of 120ml at intervals of greater than 1 month, up to a total of 960ml of blood per year. These volumes and timeframes are modest compared to those in the NHS blood donation programme where male volunteers can donate 470ml every 12 weeks (1880ml per year) and female volunteers 470ml every 16 weeks (1410ml per year). The maximum volume per donation is set at 120ml as this is the maximum volume required for our laboratory protocols and the same as that in previous, ethically approved studies in our group – in these volunteers have found 120ml to been an acceptable volume.

At the time of consent, we will ask participants to complete a brief screening questionnaire ("Case record form", final version attached). This questionnaire collects details on identifiers/ contact details (name, date of birth, email address, phone number), demographics (sex) and screening questions on relevant physical health history that may affect which experiments donated samples can be used for and interpretation of the results of laboratory experiments using the cells derived from donated blood (see section B6 for further detail).

Following recruitment/consent, volunteers will be contacted periodically (e.g. via email/phone) and invited to donate blood (as described above) for the duration of the ethics approval. At each visit details of the date and volume of blood collected will be recorded (alongside the details of any blood donations to the NHS since the last visit, if the volunteer is an NHS blood donor). This is to ensure no volunteers donate more than the study specified maximum. Volunteers can decline any individual donation or withdraw from the study at any time, both without providing a reason.

Peripheral blood leukocytes will be isolated from the blood, in the Division of Infection and Immunity research laboratories at UCL, using contemporary laboratory techniques. These will be used immediately, in cellular immunology and molecular biology laboratory experiments, or cryopreserved for future use while ethics approval remains in place. Any samples that have not been used at the end of the study will be destroyed. The experiments will study the human immune response and its interaction with microbial organisms/pathogens. This may include experiments that investigate how variation in immune responses is related to differences in host genetic code.

*Attach any questionnaires, psychological tests, etc. (a standardised questionnaire does not need to be attached, but please provide the name and details of the questionnaire together with a published reference to its prior usage).*

**B3**

**Where will the study take place (please provide name of institution/department)?**
If the study is to be carried out overseas, what steps have been taken to secure research and ethical permission in the study country?
Is the research compliant with Data Protection legislation in the country concerned or is it compliant with the General Data Protection Regulation 2018?

Division of Infection and Immunity, UCL, UK

**B4**

**Have collaborating departments whose resources will be needed been informed and agreed to participate?**
*Attach any relevant correspondence.*

Not applicable

**B5**

**How will the results be disseminated, including communication of results with research participants?**

Fully anonymised data will be published in peer-reviewed scientific journals or presented at national and international conferences. Fully anonymised gene expression/genomic data and linked phenotypic data will be made available through quality controlled public repositories to maximise their secondary use by the scientific community. Accession numbers will be provided in research papers and identifiable through conventional search facilities provided by each repository. We will not share any data from which individual patients can be identified directly or through association with other data sources.

The results of this study will have no direct impact on individual participants and we will not be providing individualised feedback on results from donated blood samples. The information obtained from donated blood samples will only be used for laboratory research- the results are not validated to make any predictions of current or future health. Our group operates a website (https://www.ucl.ac.uk/infection-immunity/research/research-department-infection/lab-research-groups/innate2adaptive) which is freely accessible to the public and via which participants can follow progress of ongoing research projects and any output. * update

**B6**

**Please outline any ethical issues that might arise from the proposed study and how they are be addressed.** *Please note that all research projects have some ethical considerations so do not leave this section blank.*

1) Recruitment, consent and withdrawal

Potential participants will include staff and students at UCL who may be acquainted with the research team. To mitigate against any perceived pressure to participate the study will be advertised indirectly e.g. via email and posters and any interested participants will self-identify themselves to the team. Thus, interaction with the research team will be initiated by potential participants. During the consent process it will be explained (verbally and via the patient information leaflet) to potential participants that any consent to participate is entirely voluntary and can be withdrawn at any time without giving a reason. The patient information leaflet also provides more than one email address via which participants can request further information or contact the study (e.g. to withdraw) in case a participant should feel reluctant to contact a given person due a prior acquaintance. The contact details provided are those of Professor Mahdad Noursadeghi (PI and applicant) and Dr Clare Thakker (Wellcome Trust Clinical Training Fellow, assisting with administration and recruitment). * update

2) Provision of personal medical history and confidentiality

At the time of recruitment, we ask volunteers to complete a brief screening questionnaire ("Case record form" attached). This questionnaire collects details on identifiers (name, date of birth, email address, phone number), demographics (sex) and screening questions on relevant physical health history (special category data). The physical health history includes questions on:

- Prior infection with/treatment for tuberculosis or vaccination against tuberculosis (BCG vaccination)
- Prior vaccination against SARS-CoV-2
- Prior infection with HIV infection, Hepatitis B/C

It is necessary to collect this data for the following reasons. We study how the human immune system interacts with the microbial organisms that cause a number of infectious diseases including tuberculosis and Covid-19. Some of our experiments investigate how immune cells from individuals that have previously been exposed to an organism (i.e. have immunological memory) respond when re-exposed, in others we want to study how cells from individuals that have no immunological memory respond. Thus, it is important to know if volunteers have had previous exposure to the organisms in question (*Mycobaterium tuberculosis*, SARS-CoV-2), either by natural infection or vaccination, in order to interpret the results and to ensure we only use blood samples from appropriate volunteers for a given experiment and not waste samples. We also ask if volunteers have history of HIV/Hepatitis B/Hepatitis C infection as these conditions/their treatments may affect immune response and alter the interpretation of results. We envisage that no-one who has been recruited will have to declare a positive history of HIV on the questionnaire as they will be excluded at the recruitment stage (HIV is a lifelong infection with lifelong

treatment and the exclusion criteria for the study include “active illness” or “regular immunomodulatory medication”). At the recruitment stage they would simply be excluded based on one these two broad reasons and would not have declare their HIV infection per se, moreover any documentary records of an exclusion criteria made at the time of enrolment will be immediately destroyed. We have opted to use screening questions rather than test for HIV in our volunteers as the UK has a low prevalence of HIV (0.15% of the population based on 2019 figures) and studies suggest 94% of those living with HIV are aware of their diagnosis. Therefore, the chance of including someone in the study who has HIV but is unaware of this is negligible and does not justify testing volunteers along with the healthcare related anxiety this may induce.

We will put in place a number of data protection measures to ensure confidentiality of personal and special category data is maintained.

All research staff participating in this work will maintain up to date Information Governance training. Study participant personal data will only be available to research staff directly associated with the study. Paper case record forms and consent forms will initially be filed in the study master file within a locked cabinet within Division of Infection and Immunity at UCL (subject to secure electronic card access). They will be scanned within 1 month after which hard copies will be shredded. The only persons to have access to this will be the PI and named members of the research team performing recruitment/ phlebotomy and thus require access.

Each participant will be allocated a unique identification number (UIN) by the recruiting research staff. The link between UIN and personal identifiers (Name and email/ phone number) will be kept separately from other personal/ special category information relating to the participants, and will only be accessible to members of the research team performing recruitment/recall in a password controlled document within an access controlled folder on UCL’s Microsoft OneDrive. Thus, all special category data will be pseudoanonymised.

At the end of analysis, samples and data will be fully anonymised by unlinking the UIN from identification information for each participant. Only fully anonymised data will be published in a relevant peer-reviewed scientific journals or presented at national and international conferences

3) Risk to participants

Phlebotomy may be associated with a risk of discomfort, bruising or bleeding. These risks are mitigated by ensuring all phlebotomists are appropriately trained. Taking a blood sample can very rarely result in fainting, and therefore we will take the blood only with the volunteer sitting in a chair or a recliner.

We judge these risks to be of low frequency and low impact. Moreover, the process of phelobotomy and associated risks will be explained during the consent process and on each visit to donate blood.

4) Incidental diagnosis of health issues and health anxiety

In accordance with the WMA Declaration of Helsinki we will not give direct feedback to participants on experiments involving donated material and this will be explained to volunteers at the time of consent.

This leads to several potential ethical issues: firstly, that the researcher may discover abnormalities in the participants samples during the course of their research which could have had health impacts for the participant and secondly, that the researcher may recruit a person for whom involvement in the research and/or the lack of specific feedback on results of blood analysis may generate health-related anxiety.

With respect to the incidental diagnosis of an underlying health problem during the course of the study several points are pertinent and make this an unlikely event. First, we anticipate our study population to be include a high number of undergraduate students, which are a group who have a very low prevalence of underlying undiagnosed medical conditions. In addition, our remaining donors are likely to predominantly be UCL staff within the Division of Infection and Immunity who are conceivably likely to be engaged with appropriate healthcare if necessary given the field in which they work. Furthermore, the analysis that we will be performing on the donated blood has not been approved or licensed for clinical diagnosis and none of the researchers are qualified clinical Biomedical Scientists who would be able to make confident diagnoses of abnormalities to a clinical standard. Indeed, very little of the testing which will be performed within our research setting has clear overlap with routine clinical testing. We will explain this to the participants as part of the consent process including that the information obtained from donated blood samples will only be used for laboratory research and the results cannot be used to make any predictions

of current or future health. All of the above minimise the risk of an ethical breach of omission as the background risk is low and we are not equipped to make clinical diagnoses in any case.

With respect to healthcare related anxiety that may be precipitated by participation in this research, we will address these issues in several ways. Firstly, in the event participants request individualised feedback on results we will explain that we are ethically unable to provide this (this will also be made explicit at enrolment). Secondly, if they are concerned regarding their underlying state of health we would advise them to see their General Practitioner.

## SECTION C DETAILS OF PARTICIPANTS

**C 1**

### Participants to be studied

| **C1a. Number of volunteers:** | 40-50 |
|---|---|
| Upper age limit: | 65 |
| Lower age limit: | 18 |

**C1b. Please justify the age range and sample size:**

We will recruit adults who are able to give autonomous written informed consent and are below the age of 65, to avoid age related confounding of immune cell biology (Ageing is believed to have an impact on immune system function, with 'immune somnolence' in the later decades of life.)

Individual donors will be invited to donate blood up to a total of 960ml of blood per year with a minimum interval of 1 month between donations and a maximum volume of 120ml per donation. Any donor may refuse to give blood on subsequent occasions without any reason.

Based on our experience, we estimate that approximately 75-100 donations per year will be required. A pool of ~20 active volunteers at any one time should be sufficient to provide this. This number allows for individual donors to decline any given invitation to donate blood e.g. due to lack of availability or personal inclination. Given that some donors will be students who will leave UCL/ the geographical area during the course of the study, and thus may wish to withdraw their consent, further volunteers may be recruited by advertisement to maintain an adequate donor pool and thus we project 40-50 volunteers will be recruited over the lifetime of the study.

**C2**

**Accessing/Using Pre-Collected Data:**
**If you are using data or information held by a third party, please explain how you will obtain this. You should confirm that the information has been obtained in accordance with the General Data Protection Regulation 2018.**

No, not applicable

**C3**

**Will the research include children or vulnerable adults such as individuals with a learning disability or cognitive impairment or individuals in a dependent or unequal relationship?** ☐ **Yes** ☒ **No**

How will you ensure that participants in these groups are competent to give consent to take part in this study? *If you have relevant correspondence, please attach it.*

Not applicable

**C4**

**Will payment or any other incentive, such as gift service or free services, be made to any research participant?**

☒ **Yes** ☐ **No** [illegible] to discuss whether this is still necessary

If yes, please specify the level of payment to be made and/or the source of the funds/gift/free service to be used.

Participant costs will be met by flat rate re-imbursement of £15. This will be offered each time they attend to donate blood. Costs will be covered by existing grant funding (Wellcome Ref: 225477/Z/22/Z, UCL Project No: 566704). Participants can accept or decline payment on each visit. To ensure accurate records and timely payment, the offer of payment, whether it was accepted and when this has been processed will be recorded by the researcher responsible for the visit alongside the other details of the blood donation (date/volume). Payments will be processed through the existing departmental finance procedures for payment of research participants.

Please justify the payment/other incentive you intend to offer.

We believe that it is important to compensate participants for costs incurred as a result of participating in the research (e.g. transport to visit the site for blood donation). This will help to ensure fair and equitable access to the study and not put off those who are less financially well-off from participating. A flat rate re-imbursement of £15 per visit should provide sufficient remuneration for travel, without further incentivising participation. A return journey within tube zone 1-5 may cost £14.60. It is anticipated that a visit to donate blood will last 15-30 minutes. Once possible travel costs are deducted, the remuneration offered is extremely unlikely to be considered an undue inducement. Furthermore, given that the risks of participation in this study are minimal so that we might expect a competent adult might accept them for no payment, the cash remuneration is very unlikely to present an undue inducement.

**C5**

## Recruitment

(i) Describe how potential participants will be identified:

The study will be advertised around UCL via the following routes

1) Poster advertising within UCL
2) Email adverts in general UCL newsletters to staff/ student email lists
3) Advertisement on the research team's website (**www.innate2adaptive.uk/volunteer**) along with a form to contact the research team
4) Posters/ flyers in UCL occupational health screening. As previously mentioned some of our experiments require cells that immunological memory to mycobacteria – in the UK, where there is a low prevalence of TB exposure/ infection, this is most often due to prior BCG vaccination. Many UCL students/staff (including all UCL Medical Students) encounter UCL Occupational Health Services where some (in particular medical students) undergo Bacillus Calmette-Guérin (BCG) vaccination, if they have not had this previously. UCL occupational health have kindly agreed to display posters/ flyers about the study. This will help ensure the posters reach a particular population of potential volunteers that are likely to have had BCG vaccination. BCG vaccination was part of the routine vaccination schedule for teenagers in the UK until 2005 and so many staff will also have received this vaccine.

These adverts will contain the inclusion/ exclusion criteria for the study. These are

- Inclusion criteria:
  - Aged 18-65 years old
- Exclusion criteria
  - Inability to give informed consent
  - Any active/ ongoing health conditions/illness
  - Any regular immunomodulatory medication

Given the advertisement will be within UCL, we anticipate that the vast majority of our participants will be current UCL staff and students.

Any potential volunteers who identify themselves as eligible and make contact with the study team will be provided with patient information leaflets. They will then be given as much time as they wish to consider the invitation to participate, with the opportunity to ask additional questions. Those who agree to participate will be required to provide written informed consent.

(ii) Describe how potential participants will be approached:

As described above potential volunteers will self-identify themselves in response to the advertisements around UCL and, should they wish to participate/ received further information, approach the study team using the contact details provided on the advertisements.

(iii) Describe how participants will be recruited:

Once a potential volunteer makes contact with the study team, the researcher will describe the study to them, provide them with the participant information sheet and answer their questions. They will then be given as much time as they wish to consider the information, with the opportunity to ask additional questions. Those who agree to participate will be required to provide written informed consent. As we anticipate the vast majority of participants will be UCL staff/ students, provision will not need to be made for non-English speakers. Participants who have consented to take part in the study will be contacted and recalled to donate blood periodically (but not exceeding a frequency of once a month). Volunteers can decline any individual donation or withdraw from the study at any time without giving a reason.

**C6**

**Will the participants participate on a fully voluntary basis?** ☒ **Yes** ☐ **No**

**Will UCL students be involved as participants in the research project?** ☒ **Yes** ☐ **No**

*If yes, care must be taken to ensure that they are recruited in such a way that they do not feel any obligation to a teacher or member of staff to participate.*

**Please state how you will bring to the attention of the participants their right to withdraw from the study without penalty?**

Participants can withdraw at any time without giving a reason and without penalty. They will be advised of this both verbally on first meeting the research team and in writing, on the Information Sheet and Consent Form.

To protect students from a feeling of obligation to participate we will take the precautions outlined in Section B6. In particular, the study will be advertised indirectly e.g. via email and posters and any interested participants will self-identify themselves to the team. Thus, interaction with the research team will be initiated by potential participants. During the consent process it will be explained (verbally and via the patient information leaflet) to potential participants that any consent to participate is entirely voluntary and can be withdrawn at any time without giving a reason. The patient information leaflet also provides more than one email address via which participants can request further information or contact the study (e.g. to withdraw) in case a participant should feel reluctant to contact a given person due a prior acquaintance.

**C7**

## CONSENT

**Please describe the process you will use when seeking and obtaining consent.**

Once a potential participant contacts the study team, in response to the advertisements described above, the researcher will describe the study to potential participants, provide them with the participant information sheet and answer their questions. The description of the study will include a description of the phlebotomy process, explained by a trained phlebotomist who is a member of laboratory staff. The researcher will explicitly advise the participant at this time that they may withdraw their consent at any point without giving a reason.

The researcher will then proceed to confirm understanding from the participant of the stated aims and the procedure for phlebotomy and answer any questions the participant may have. The participant will then be given as much time as they wish to consider if they wish to take part, with the opportunity to ask additional questions.

If the participant wishes to take part, they will then be asked to sign a consent form which will be held with the study records securely and a copy will be provided to the participant.

*A copy of your participant information sheet(s) and consent form(s) must be attached to this application. For your convenience proformas are provided in Appendix I. These should be filled in and modified as necessary.*

In cases where it is not proposed to obtain the participants informed consent, please explain why below.

Not applicable

| | |
|---|---|
| | |

| | |
|---|---|
| **C8** | **Will any form of deception be used that raises ethical issues? If so, please explain.**<br><br>No |

| | |
|---|---|
| **C9** | **Will you provide a full debriefing at the end of the data collection phase?** ☐ **Yes** ☒ **No**<br>If 'No', please explain why below.<br><br>At the end of each phlebotomy visit the researcher will check that the participant feels well and will answer any questions they have.<br><br>The results of this study will have no direct impact on individual participants and we will not be providing individualised feedback on results from donated blood samples. The information obtained from donated blood samples will only be used for laboratory research- the results cannot be used to make any predictions of current or future health. Our group operates a website (https://www.ucl.ac.uk/infection-immunity/research/research-department-infection/lab-research-groups/innate2adaptive) which is freely accessible to the public and via which participants can follow progress of ongoing research projects and any output. |

| | |
|---|---|
| **C10** | **Information Sheets And Consent Forms: Appendix I**<br><br>**A poorly written Information Sheet(s) and Consent Form(s) that lack clarity and simplicity frequently delay ethics approval of research projects.** The wording and content of the Information Sheet and Consent Form must be appropriate to the age and educational level of the research participants and clearly state in simple non-technical language what the participant is agreeing to. Use the active voice e.g. "we will book" rather than "bookings will be made". Refer to participants as "you" and yourself as "I" or "we". An appropriate translation of the Forms should be provided where the first language of the participants is not English. If you have different participant groups you should provide Information Sheets and Consent Forms as appropriate (e.g. one for children and one for parents/guardians) using the templates provided in Appendix I. Where children are of a reading age, a written Information Sheet should be provided. When participants cannot read or the use of forms would be inappropriate, a description of the verbal information to be provided should be given. Where possible please ensure that you trial the forms on an age-appropriate person before you submit your application. |

## SECTION D: APPROPRIATE SAFEGUARDS, DATA STORAGE AND SECURITY

| | |
|---|---|
| **D1** | **Will the research involve the collection and/or use of personal data?**<br>☒ **Yes** ☐ **No**<br><br>***Personal data*** *is data which relates to a living individual who can be identified from that data OR from the data and other information that is either currently held, or will be held by the data controller (the researcher).* |

*This includes:*

- *any expression of opinion about the individual and any intentions of the data controller or any other person toward the individual.*
- *sensor, location or visual data which may reveal information that enables the identification of a face, address, etc (some postcodes cover only one property).*
- *combinations of data which may reveal identifiable data, such as names, email/postal addresses, date of birth, ethnicity, descriptions of health diagnosis or conditions, computer IP address (if relating to a device with a single user).*

**If yes, is the research collecting or using special category data as defined by the GDPR 2018**, for example data:

- which reveals racial or ethnic origin, political opinions, religious or philosophical beliefs, trade union membership ;
- data concerning health (the physical or mental health of a person, including the provision of health care services) ;
- data concerning sex life or sexual orientation ; or
- genetic or biometric data processed to uniquely identify a natural person.

- **data which might be considered sensitive in some countries, cultures or contexts?**

**Note that if you intend to process 'special category' information you will need an 'additional' legal basis for processing that particular data and further safeguards will need to be put in place.**

**If yes, state whether explicit ethical informed consent will be sought for its use and what data management measures are in place to adequately manage and protect the data.**

The research is collecting data on medical physical health- at the time of enrolment we will ask participants to complete a brief screening questionnaire ("Case record form" attached). This questionnaire collects details on identifiers (name, date of birth, email address, phone number), demographics (sex) and screening questions on relevant physical health history that may affect which experiments donated samples can be used for.

The legal basis used to process special category personal data will be for scientific research. This is detailed in the patient information leaflet.

Explicit ethical informed consent will be sought. The patient information leaflet (PIL) explains what data will be collected on the questionnaire and why we are collecting this data (PIL attached, page 2). This will also be explained verbally to participants at the time of consent and the consent form seeks explicit consent to collect personal health information (see attached consent form, item 4)

The data management plan is detailed below in D2.

**D2**

**During the Project (including the write up and dissemination period)**

**State what types of data will be generated from this project** (i.e. transcripts, videos, photos, audio tapes, field notes, etc).

For each of the enrolled participants, we will collect data on identifiers (name, date of birth, email address, phone number), demographics (sex) and screening questions on relevant physical health history ("Case record form" attached).

Blood samples will be pseudoanonymised and used immediately, in cellular immunology and molecular biology laboratory experiments to investigate the factors which make them more or less susceptible to infection and how they react to or are otherwise affected by infection (this may include genotyping and studies of gene expression), or cryopreserved for future use within the time frame of this ethics approval.

**How will data be stored, including where and for how long?** This includes all hard copy and electronic data on laptops, share drives, usb/mobile devices.

At the time of consent, participants will complete a case record form giving details of identifiers (name, date of birth, email address, phone number), demographics (sex) and screening questions on relevant physical health history that may affect which experiments donated samples can be used for ("Case record form" attached).

Paper case record forms and consent forms will initially be filed in the study master file within a locked cabinet within Division of Infection and Immunity at UCL (subject to secure electronic card access). Hard copy paper records will be scanned within 1-month (and subsequently stored within an access controlled folder on UCL's Microsoft OneDrive) and hard copies shredded at the time of scanning. The only persons to have access to this will be the PI and named members of the research team performing recruitment/ phlebotomy.

Each participant will be allocated a unique identification number (UIN) by the recruiting research staff. The link between UIN and personal identifiers (Name and email/ phone number) will be kept separately from all other information relating to the participants, and will only be accessible to members of the research team

performing recruitment/recall in a password controlled document within an access controlled folder on UCL's Microsoft OneDrive.

An electronic log of participants will be stored in a password controlled document on Microsoft OneDrive within an access controlled folder (accessible only to the research team). The electronic log book will be psuedoanoymised (identified by UINs only) and record relevant physical medical history obtained on the screening questionnaire, details of dates and volume of blood donations (to ensure no volunteers donate more than the study specified maximum). At the end of analysis, all data will be fully anonymised by unlinking the UIN from identification information for each participant.

Blood samples will be pseudanonymised, through use of the UIN. At the end of analysis, samples will be fully anonymised by unlinking the UIN from identification information for each participant. Peripheral blood leukocytes will be isolated from the blood using contemporary laboratory techniques in the Division of Infection and Immunity research laboratories at UCL. The majority of these will be used immediately, in cellular immunology and molecular biology laboratory experiments (this may include genotyping and studies of gene expression). Some samples may be cryopreserved for future use within the time frame of this ethics approval. Cryopreserved samples will be stored on HTA licensed premises at UCL in accordance with the Human Tissue Act (2004). Any samples that have not been used at the end of the study's ethics approval will be destroyed. Data resulting from laboratory experiments (anonymised) will be stored on Microsoft OneDrive, UCL Research Data Storage (RDS) Service (e.g. transcriptomic data) or password protected portable hard drives.

Only fully anonymised data will be published in a relevant peer-reviewed journals or presented at national and international conferences. Fully anonymised gene expression/genomic data and linked phenotypic data will be made available through quality controlled public repositories to maximise their use by the scientific community. We will not share any data from which individual patients can be identified directly or through association with other data sources.

No identifiable data will be shared outside the laboratory or the institution.

If email is used to invited consented participants to donate blood, this will be via individual email or BCC emails to ensure anonymity and privacy is maintained.

**Who will have access to the data, including advisory groups and during transcription?**

Professor Noursadeghi will be the custodian of all personal data and laboratory data for each participant. Personal data will only be available to research staff directly associated with the study.

**D3**

**Will personal data be processed or be sent outside of the European Economic Area (EEA)*?**

**If yes,** please confirm that there are adequate levels of protection in compliance with the General Data Protection Regulation 2018 and state what these arrangements are below.

No

**D 4**

**After the Project**

**What data will be stored and how will you keep it secure?**

The only data to be stored after the study will be fully anonymised processed outputs of biological analysis and associated fully anonymised metadata e.g. vaccine status of the donors for samples used in a given experiment. These may include anonymised data shown in graphs, tables and diagrams and raw numerical and graphical data. The link between UIN and personal identifiers will be destroyed at the end of the study, rendering all experimental data listed above fully anonymised. No personal identifiable information will be stored beyond the duration of the study.

**Where will the data be stored and who will have access?**

Fully anonymised research data will be stored electronically in

- UCL computer facilities (maintained by information technology services within the Division of Infection & Immunity at UCL) and backed up by automated synchronisation within dedicated divisional servers with password protected access
- Encrypted/password protected portable media including hard drives and removable disc

This fully anonymised research data will continue to be used by research staff within the Noursadeghi research group and for presentation and publication after the termination of the study.

Only fully anonymised data will be published in a relevant peer-reviewed journals or presented at national and international conferences. Fully anonymised gene expression/genomic data and linked phenotypic data will be made available through quality controlled public repositories to maximise their use by the scientific community. We will not share any data from which individual patients can be identified directly or through association with other data sources.

**Will the data be securely deleted?**

**If yes,** please state when will this occur:

At the end of the study all personal data will be deleted. Anonymised results and outputs from research will be retained as described above.

**D 5**

**Will the data be archived for use by other researchers?** ☒ **Yes** ☐ **No**

If **Yes**, please describe provide further details including whether researchers outside the EEA will be given access.

This fully anonymised research data will continue to be used by research staff within the Noursadeghi research group and for presentation and publication after the termination of the study.

Processed/analysed anonymised data sets will also be made available through supplementary on-line content associated with peer-reviewed scientific publications.

Fully anonymised gene expression/genomic data and linked phenotypic data will be made available through quality controlled public repositories to maximise their use by the scientific community. We will not share any data from which individual patients can be identified directly or through association with other data sources.

## SECTION E: DETAILS OF RISKS AND BENEFITS TO THE RESEARCHER AND THE RESEARCHED

**E1**

**Please state briefly any precautions being taken to protect the health and safety of researchers and others associated with the project (as distinct from the research participants).**

Phlebotomy carried a risk of sharps injury to the practitioner. This risk is mitigated by phlebotomy training for all staff taking blood and the provision of a dedicated phlebotomy room with appropriate facilities for sharps disposal.

Standardised laboratory protocols and risk assessments for processing of blood samples and experimental work in the research laboratory are in place.

| | |
|---|---|
| **E 2** | **Will these participants participate in any activities that may be potentially stressful or harmful in connection with this research?**<br>☒ **Yes** ☐ **No**<br><br>If **Yes**, please describe the nature of the risk or stress and how you will minimise and monitor it.<br>Phlebotomy may be associated with a risk of discomfort, bruising or bleeding. These risks are mitigated by ensuring all phlebotomists are appropriately trained. Taking a blood sample can very rarely result in fainting, and therefore we will take the blood only with the volunteer sitting in a chair or a recliner. We judge these risks to be of low frequency and low impact. Moreover, the process of phlebotomy and associated risks will be explained during the consent process and on each visit to donate blood. |

| | |
|---|---|
| **E3** | **Will group or individual interviews/questionnaires raise any topics or issues that might be sensitive, embarrassing or upsetting for participants?**<br><br>If **Yes,** please explain how you will deal with this.<br><br>At the time of recruitment, we ask volunteers to complete a brief screening questionnaire ("Case record form" attached). This questionnaire collects details on identifiers (name, date of birth, email address, phone number), demographics (sex) and screening questions on relevant physical health history (special category data). The physical health history include questions on:<br>• Prior infection with/treatment for tuberculosis or vaccination against tuberculosis (BCG vaccination)<br>• Prior vaccination against SARS-CoV-2<br>• Prior infection with HIV infection, Hepatitis B/C<br><br>It is necessary to collect this data for the following reasons. We study how the human immune system interacts with the microbial organisms that cause a number of infectious diseases including tuberculosis and Covid-19. Some of our experiments investigate how immune cells from individuals that have previously been exposed to an organism (i.e. have immunological memory) respond when re-exposed, in others we want to study how cells from individuals that have no immunological memory respond. Thus, it is important to know if volunteers have had previous exposure to the organisms in question (Mycobacterium tuberculosis, SARS-CoV-2), either by natural infection or vaccination, in order to interpret the results and to ensure we only use blood samples from appropriate volunteers for a given experiment and not waste samples. We also ask if volunteers have history of HIV/Hepatitis B/Hepatitis C infection as these conditions/their treatments may affect immune response and alter the interpretation of results. We envisage that no-one who has been recruited will have to declare a positive history of HIV on the questionnaire as they will be excluded at the recruitment stage (HIV is a lifelong infection with lifelong treatment and the exclusion criteria for the study include "active illness" or "regular immunomodulatory medication"). At the recruitment stage they would simply be excluded based on one these broad reasons and would not have declare their HIV infection per se, moreover any documentary records of an exclusion criteria made at the time of enrolment will be immediately destroyed. We have opted to use screening questions rather than test for HIV in our volunteers as the UK has a low prevalence of HIV (0.15% of the population based on 2019 figures) and studies suggest 94% of those living with HIV are aware of their diagnosis. Therefore, the risk of including someone in the study who has HIV but is unaware of this is negligible and does not justify testing volunteers along with the healthcare related anxiety this may induce.<br><br>All information provided will be kept strictly confidential (as described in the data management plan, section D2) and this will be explained to participants. |
| **E4** | **Please describe any expected benefits to the participant.**<br>There are no immediate benefits for those people participating in the project. The patient information leaflet explains this and the fact that volunteers will be making a contribution to science. |

**E5**

**Specify whether the following procedures are involved:**

**Any invasive procedure(s)** ☒ **Yes** ☐ **No**

**Physical contact** ☒ **Yes** ☐ **No**

**Any procedure(s) that may cause mental distress** ☐ **Yes** ☒ **No**

Please state briefly any precautions being taken to protect the health and safety of the research participants.

Phlebotomy may be associated with a risk of discomfort, bruising or bleeding. These risks are mitigated by ensuring all phlebotomists are appropriately trained. Taking a blood sample can very rarely result in fainting, and therefore we will take the blood only with the volunteer sitting in a chair or a recliner. We judge these risks to be of low frequency and low impact. Moreover, the process of phlebotomy and associated risks will be explained during the consent process and on each visit to donate blood.

**E6**

**Does the research involve the use of drugs?** ☐ **Yes** ☒ **No**

If **Yes**, please name the drug/product and its intended use in the research and then complete Appendix II

Not applicable

**Does the project involve the use of genetically modified materials?** ☐ **Yes** ☒ **No**

If **Yes**, has approval from the Genetic Modification Safety Committee been obtained for work? ☐ Yes ☐ No

If **Yes**, please quote the Genetic Modification Reference Number: Not applicable

**E7**

**Will any non-ionising radiation be used on the research participant(s)?** ☐ **Yes** ☒ **No**

If **Yes**, please complete Appendix III.

**E8**

**Are you using a medical device in the UK that is CE-marked and is being used within its product indication?** ☐ **Yes** ☒ **No**

If **Yes**, please complete Appendix IV.

## CHECKLIST

| Documents to be Attached to Application Form (if applicable) | Tick if attached |
|---|---|
| **Section B: Details of the Project** | |
| • Questionnaire(s) / Psychological Tests "**Participant case record form"** | ☒ |
| • Relevant correspondence relating to involvement of collaborating department/s and agreed participation in the research i.e. approval letters to gatekeepers seeking permission to do research on their premises/ in their company etc. | ☐ |
| **Section C: Details of Participants** | |
| • Parental/guardian consent form for research involving participants under 18 | ☐ |
| • Participant/s information sheet | ☒ |

- Participant/s consent form/s ☒
- Advertisement ☒

**Appendix I: Information Sheet(s) and Consent Form(s)** ☒

**Appendix II: Research Involving the Use of Drugs**

- Relevant correspondence relating to agreed arrangements for dispensing with the pharmacy ☐
- Written confirmation from the manufacturer that the drug/substance has has been manufactured to GMP ☐
- Proposed volunteer contract ☐
- Full declaration of financial or direct interest ☐
- Copies of certificates: CTA etc… ☐

**Appendix III: Use of Non-Ionising Radiation** ☐

**Appendix IV: Use of Medical Devices** ☐

Updated October 2019